# Basics of Data Science

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## Homework 0 Solutions

**Topics:** Supervised Learning Foundations, Bias-Variance Decomposition, Augmented Ridge Regression, Softmax Concavity, and Cross-Validation Leakage

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## Problem 1: Expected Prediction Error & Bias-Variance in $k$-NN [25 points]

**Problem Statement:** Consider the classical nonparametric regression setting. Suppose we have a training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ where the scalar targets are generated by the model $y = f(x) + \epsilon$, with conditional zero mean $\mathbb{E}[\epsilon \mid x] = 0$ and homoscedastic variance $\mathrm{Var}(\epsilon \mid x) = \sigma^2$. The additive noise $\epsilon$ is mutually independent across observations and independent of the input vector $x \in \mathbb{R}^p$.

For a fixed query test point $x_0 \in \mathbb{R}^p$:

(a) Show that under squared-error loss, the conditional Expected Prediction Error of the 1-NN estimator $\hat{f}_1(x_0) = y_{(1)}$ decomposes into irreducible noise, squared bias, and estimator variance.

(b) Generalize the decomposition to the $k$-NN estimator $\hat{f}_k(x_0) = \frac{1}{k}\sum_{\ell=1}^k y_{(\ell)}$.

(c) Analyze the trade-off as $k$ varies from 1 to $N$, and explain why bias fails to vanish in high dimensions ($p \gg 1$).

### Solution:

**Part (a): 1-Nearest-Neighbor Decomposition [9 points]**

Let $(x_0, y_0)$ be an independent out-of-sample query observation with response $y_0 = f(x_0) + \epsilon_0$, where $\epsilon_0$ is independent of the training data $\mathcal{D}$ and satisfies $\mathbb{E}[\epsilon_0] = 0$ and $\mathrm{Var}(\epsilon_0) = \sigma^2$.

The 1-Nearest-Neighbor estimator evaluated at $x_0$ is:

$$\hat{f}_1(x_0) = y_{(1)} = f(x_{(1)}) + \epsilon_{(1)},$$

where $x_{(1)} \in \{x_1, \ldots, x_N\}$ denotes the training input closest to $x_0$ in Euclidean distance, and $\epsilon_{(1)}$ is its corresponding training observation noise.

Subtracting the estimator from the test target:

$$y_0 - \hat{f}_1(x_0) = (f(x_0) + \epsilon_0) - \Big(f(x_{(1)}) + \epsilon_{(1)}\Big) = \Big[f(x_0) - f(x_{(1)})\Big] + \epsilon_0 - \epsilon_{(1)}.$$

Now we square this error and take the conditional expectation with respect to the test noise $\epsilon_0$ and the training targets conditioned on the fixed sample locations $\{x_i\}_{i=1}^N$ and query point $x_0$:

$$\begin{aligned}
\mathrm{EPE}(x_0) &\equiv \mathbb{E}\left[(y_0 - \hat{f}_1(x_0))^2 \;\middle|\; x_0, \{x_i\}_{i=1}^N\right] \\
&= \mathbb{E}\left[\Big([f(x_0) - f(x_{(1)})] + (\epsilon_0 - \epsilon_{(1)})\Big)^2 \;\middle|\; x_0, \{x_i\}_{i=1}^N\right] \\
&= \Big[f(x_0) - f(x_{(1)})\Big]^2 + \mathbb{E}[\epsilon_0^2] + \mathbb{E}[\epsilon_{(1)}^2] \\
&\quad + 2\Big[f(x_0) - f(x_{(1)})\Big]\underbrace{\mathbb{E}[\epsilon_0]}_{=0} - 2\Big[f(x_0) - f(x_{(1)})\Big]\underbrace{\mathbb{E}[\epsilon_{(1)}]}_{=0} - 2\underbrace{\mathbb{E}[\epsilon_0\epsilon_{(1)}]}_{=0}.
\end{aligned}$$

Because $\epsilon_0$ and $\epsilon_{(1)}$ are independent zero-mean random variables with $\mathbb{E}[\epsilon_0^2] = \sigma^2$ and $\mathbb{E}[\epsilon_{(1)}^2] = \sigma^2$, all interaction cross-terms vanish identically:

$$\text{EPE}(x_0) = \underbrace{\sigma^2}_{\text{Irreducible Error } \text{Var}(\epsilon_0)} + \underbrace{\Big[f(x_0) - f(x_{(1)})\Big]^2}_{\text{Squared Bias } [\text{Bias}(\hat{f}_1(x_0))]^2} + \underbrace{\sigma^2}_{\text{Sampling Variance } \text{Var}(\hat{f}_1(x_0))}. \quad \blacksquare$$

The three fundamental components are:

- **Irreducible Noise ($\sigma^2$):** The intrinsic variation in the target $y_0$ that no model, regardless of complexity, can predict.
- **Squared Model Bias ($[f(x_0) - f(x_{(1)})]^2$):** The systematic geometric approximation error arising from using the true function value at the closest training sample $x_{(1)}$ instead of the target point $x_0$.
- **Estimator Variance ($\sigma^2$):** The sampling variability of the estimator induced by the observation noise $\epsilon_{(1)}$ at the selected neighbor.

**Part (b): Generalization to $k$-Nearest Neighbors [9 points]**
For the $k$-NN averaging estimator:

$$\hat{f}_k(x_0) = \frac{1}{k}\sum_{\ell=1}^{k} y_{(\ell)} = \frac{1}{k}\sum_{\ell=1}^{k} f(x_{(\ell)}) + \frac{1}{k}\sum_{\ell=1}^{k} \epsilon_{(\ell)}.$$

Conditioning on the fixed positions $\{x_i\}_{i=1}^N$:

1. **Conditional Expectation:**

$$\mathbb{E}\left[\hat{f}_k(x_0) \;\middle|\; \{x_i\}\right] = \frac{1}{k}\sum_{\ell=1}^{k} f(x_{(\ell)}) + \frac{1}{k}\sum_{\ell=1}^{k} \underbrace{\mathbb{E}[\epsilon_{(\ell)}]}_{=0} = \frac{1}{k}\sum_{\ell=1}^{k} f(x_{(\ell)}).$$

2. **Squared Bias:**

$$\text{Bias}\left(\hat{f}_k(x_0)\right) = \mathbb{E}\left[\hat{f}_k(x_0)\right] - f(x_0) = \left(\frac{1}{k}\sum_{\ell=1}^{k} f(x_{(\ell)})\right) - f(x_0).$$

   Squaring this quantity yields $\left[f(x_0) - \frac{1}{k}\sum_{\ell=1}^{k} f(x_{(\ell)})\right]^2$.

3. **Estimator Variance:** Since the noise terms $\epsilon_{(\ell)}$ are mutually independent with common variance $\sigma^2$:

$$\text{Var}\left(\hat{f}_k(x_0) \;\middle|\; \{x_i\}\right) = \text{Var}\left(\frac{1}{k}\sum_{\ell=1}^{k} \epsilon_{(\ell)}\right) = \frac{1}{k^2}\sum_{\ell=1}^{k} \text{Var}(\epsilon_{(\ell)}) = \frac{1}{k^2}(k\sigma^2) = \frac{\sigma^2}{k}.$$

Summing the irreducible noise, squared bias, and variance gives the exact analytical formula:

$$\text{EPE}(x_0; k) = \sigma^2 + \left[f(x_0) - \frac{1}{k}\sum_{\ell=1}^{k} f(x_{(\ell)})\right]^2 + \frac{\sigma^2}{k}. \quad \blacksquare$$

**Part (c): Complexity Trade-Off and the Curse of Dimensionality [7 points]**

- **Extreme $k = 1$ (High Model Complexity / Overfitting):** Variance achieves its highest value $\text{Var} = \sigma^2$. The model fits local training noise aggressively. However, bias is minimized because only the closest available training point $x_{(1)}$ is queried.

- **Extreme $k = N$ (Low Model Complexity / Underfitting):** Variance is suppressed to its minimum $\text{Var} = \frac{\sigma^2}{N} \approx 0$. However, the estimator reduces to predicting the global training sample average $\bar{y}$, making the bias $\left[f(x_0) - \frac{1}{N}\sum_{i=1}^{N} f(x_i)\right]^2$ very large whenever $f$ is non-constant.
- **The Curse of Dimensionality in High Dimensions ($p \gg 1$):** In $\mathbb{R}^p$, the volume of a hypercube of side length $d \in (0, 1)$ scales as $d^p$. To capture a fixed fraction $r$ of the data points, the neighborhood diameter must scale as $d \approx r^{1/p}$. As $p$ grows large, $r^{1/p} \to 1$ even for small fractions $r$ (e.g., for $r = 0.01$ and $p = 20$, $d = 0.01^{1/20} \approx 0.794$). Consequently, all training points become equidistant and lie on the boundary of the space. Even the closest neighbor $x_{(1)}$ is geometrically far from $x_0$, so $|f(x_0) - f(x_{(1)})| \not\to 0$. The bias term fails to vanish, causing nonparametric nearest-neighbor models to break down without severe dimension reduction or structural regularization. ■

## Problem 2: Ridge Regression as an Augmented OLS System [25 points]

**Problem Statement:** In high-dimensional and collinear data regimes, Ordinary Least Squares can suffer from severe variance inflation or non-invertibility of $X^T X$. Ridge regression stabilizes parameter estimates by penalizing the $L_2$-norm of the regression coefficients:

$$\min_{\beta \in \mathbb{R}^p} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}, \quad (\lambda > 0).$$

Define the augmented feature matrix $X^* = \begin{bmatrix} X \\ \sqrt{\lambda} I_p \end{bmatrix}$ and augmented response $y^* = \begin{bmatrix} y \\ 0_p \end{bmatrix}$.

(a) Define the augmented data system $(X^*, y^*)$.

(b) Show algebraically that the augmented OLS problem $\min_\beta \|y^* - X^*\beta\|_2^2$ is identically equal to the Ridge regression objective.

(c) Show that the closed-form augmented OLS solution reproduces $\hat{\beta}_{\text{ridge}} = (X^T X + \lambda I_p)^{-1} X^T y$.

(d) Prove that $X^{*T} X^*$ is strictly positive definite for any $\lambda > 0$, guaranteeing invertibility even when $N < p$ or when columns of $X$ are collinear.

### Solution:

**Part (a): Augmented Data System Definition [5 points]**
We define the vertically stacked augmented feature matrix $X^* \in \mathbb{R}^{(N+p)\times p}$ and response vector $y^* \in \mathbb{R}^{N+p}$ as:

$$X^* = \begin{bmatrix} X \\ \sqrt{\lambda} I_p \end{bmatrix}, \quad y^* = \begin{bmatrix} y \\ 0_p \end{bmatrix},$$

where $I_p$ denotes the $p \times p$ identity matrix and $0_p$ is a column vector of $p$ zeros. The first $N$ rows correspond to the original empirical data, while the bottom $p$ rows represent synthetic virtual observations penalizing deviation from zero.

**Part (b): Equivalence of the Objectives [6 points]**
Compute the augmented residual vector by matrix-vector block arithmetic:

$$y^* - X^*\beta = \begin{bmatrix} y \\ 0_p \end{bmatrix} - \begin{bmatrix} X \\ \sqrt{\lambda} I_p \end{bmatrix} \beta = \begin{bmatrix} y - X\beta \\ 0_p - \sqrt{\lambda}\beta \end{bmatrix} = \begin{bmatrix} y - X\beta \\ -\sqrt{\lambda}\beta \end{bmatrix}.$$

The squared Euclidean norm of a partitioned vector is the sum of the squared Euclidean norms of its subvectors:

$$\begin{aligned}\|y^* - X^*\beta\|_2^2 &= \left\| \begin{bmatrix} y - X\beta \\ -\sqrt{\lambda}\beta \end{bmatrix} \right\|_2^2 = \|y - X\beta\|_2^2 + \| - \sqrt{\lambda}\beta\|_2^2 \\ &= \|y - X\beta\|_2^2 + (\sqrt{\lambda})^2\|\beta\|_2^2 \\ &= \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2.\end{aligned}$$

This proves that the standard ordinary least squares objective on the augmented dataset $(X^*, y^*)$ is identically equal to the penalized Ridge regression objective for any candidate $\beta \in \mathbb{R}^p$. ■

**Part (c): Equivalence of Closed-Form Estimators [7 points]**
Compute the Gram matrix of the augmented feature matrix $X^*$:

$$X^{*T}X^* = \begin{bmatrix} X^T & \sqrt{\lambda}I_p \end{bmatrix} \begin{bmatrix} X \\ \sqrt{\lambda}I_p \end{bmatrix} = X^TX + (\sqrt{\lambda}I_p)(\sqrt{\lambda}I_p) = X^TX + \lambda I_p.$$

Next, compute the cross-product vector between $X^*$ and $y^*$:

$$X^{*T}y^* = \begin{bmatrix} X^T & \sqrt{\lambda}I_p \end{bmatrix} \begin{bmatrix} y \\ 0_p \end{bmatrix} = X^Ty + \sqrt{\lambda}I_p(0_p) = X^Ty.$$

The normal equations for the unconstrained augmented OLS problem state that $\hat{\beta}^*$ must satisfy:

$$(X^{*T}X^*)\hat{\beta}^* = X^{*T}y^* \quad \iff \quad (X^TX + \lambda I_p)\hat{\beta}^* = X^Ty.$$

Assuming invertibility (proven in Part (d)), solving for $\hat{\beta}^*$ gives:

$$\hat{\beta}^* = (X^{*T}X^*)^{-1}X^{*T}y^* = (X^TX + \lambda I_p)^{-1}X^Ty = \hat{\beta}_{\text{ridge}}. \quad ■$$

**Part (d): Strict Positive Definiteness and Numerical Invertibility [7 points]**
Let $v \in \mathbb{R}^p \setminus \{0_p\}$ be any non-zero vector. Evaluate the quadratic form associated with $X^{*T}X^*$:

$$v^T(X^{*T}X^*)v = v^T(X^TX + \lambda I_p)v = v^TX^TXv + \lambda v^Tv = \|Xv\|_2^2 + \lambda\|v\|_2^2.$$

- Since the Euclidean norm is always non-negative, $\|Xv\|_2^2 \geq 0$.
- Since $v \neq 0_p$, its squared norm is strictly positive: $\|v\|_2^2 > 0$.
- Since the regularization parameter $\lambda > 0$, the product $\lambda\|v\|_2^2 > 0$.

Therefore, we have the strict inequality:

$$v^T(X^{*T}X^*)v = \|Xv\|_2^2 + \lambda\|v\|_2^2 \geq 0 + \lambda\|v\|_2^2 > 0 \quad \forall v \neq 0_p.$$

This proves that $X^{*T}X^* = X^TX + \lambda I_p$ is **strictly positive definite** for any $\lambda > 0$.

**Implications for Machine Learning and Ill-Posed Systems:**

- **Case $N < p$ (High-Dimensional Regime):** The raw Gram matrix $X^TX$ has rank at most $N < p$, meaning it has at least $p - N$ zero eigenvalues and is completely non-invertible in standard OLS. By adding $\lambda I_p$, all eigenvalues $\mu_j(X^TX)$ are shifted upward by $\lambda$:

  $$\mu_j(X^TX + \lambda I_p) = \mu_j(X^TX) + \lambda \geq 0 + \lambda = \lambda > 0.$$

  Thus, zero eigenvalues are eliminated, guaranteeing a unique, well-defined solution exists even when $p \gg N$.

- **Multicollinearity & Numerical Conditioning:** When features are highly correlated, the condition number $\kappa(X^TX) = \frac{\mu_{\max}}{\mu_{\min}} \to \infty$, leading to catastrophic numerical error and massive coefficient variance. Ridge regularization bounds the condition number by $\kappa(X^TX + \lambda I_p) \leq \frac{\mu_{\max}+\lambda}{\lambda} < \infty$, stabilizing the system against matrix inversion perturbations. ■

## Problem 3: Multiclass Softmax Regression & Log-Likelihood Concavity[25 points]

**Problem Statement:** Consider a multiclass classification problem with $K \geq 2$ classes and 1-of-$K$ target encoding $y_i = [y_{i1}, \ldots, y_{iK}]^T \in \{0,1\}^K$ with $\sum_{k=1}^K y_{ik} = 1$. The class posterior probabilities follow the Softmax formulation:

$$p_k(x_i;\theta) \equiv P(Y = k \mid x_i) = \frac{\exp(\beta_k^T x_i)}{\sum_{\ell=1}^K \exp(\beta_\ell^T x_i)}, \quad k = 1, \ldots, K.$$

(a) Formulate the conditional log-likelihood function $\ell(\theta) = \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log p_k(x_i;\theta)$.

(b) Derive the score gradient $\nabla_{\beta_k} \ell(\theta) = \sum_{i=1}^N (y_{ik} - p_k(x_i;\theta))x_i$ and interpret it.

(c) Compute the block Hessian $\mathcal{H}_{k\ell} = -\sum_{i=1}^N p_k(x_i)(\delta_{k\ell} - p_\ell(x_i))x_i x_i^T$.

(d) Prove that $-\mathcal{H} \succeq 0$, establishing global concavity and uniqueness of the maximum likelihood estimate.

### Solution:

**Part (a): Formulation of the Conditional Log-Likelihood [5 points]**
For an independent sample $i$, the response vector $y_i$ is governed by a single-trial multinomial distribution with event probabilities $p_1(x_i), \ldots, p_K(x_i)$. The probability of observation $(x_i, y_i)$ is:

$$P(y_i \mid x_i;\theta) = \prod_{k=1}^K [p_k(x_i;\theta)]^{y_{ik}}.$$

Taking the product over all $N$ independent observations in $\mathcal{D}$ and computing the natural logarithm yields the total log-likelihood function:

$$\ell(\theta) = \ln \prod_{i=1}^N \prod_{k=1}^K [p_k(x_i;\theta)]^{y_{ik}} = \sum_{i=1}^N \sum_{k=1}^K y_{ik} \ln p_k(x_i;\theta). \quad ■$$

**Part (b): Score Equations and Physical Interpretation [7 points]**
Define the linear predictor score $\eta_{ik} = \beta_k^T x_i$. By the chain rule, for any class $m \in \{1, \ldots, K\}$:

$$\frac{\partial p_m(x_i)}{\partial \eta_{ik}} = \frac{\partial}{\partial \eta_{ik}} \left[ \frac{\exp(\eta_{im})}{\sum_{\ell=1}^K \exp(\eta_{i\ell})} \right].$$

Using the quotient rule:

- If $m = k$: $\frac{\partial p_k(x_i)}{\partial \eta_{ik}} = \frac{\exp(\eta_{ik}) \sum_\ell \exp(\eta_{i\ell}) - \exp(\eta_{ik}) \exp(\eta_{ik})}{\left(\sum_\ell \exp(\eta_{i\ell})\right)^2} = p_k(x_i)(1 - p_k(x_i))$.
- If $m \neq k$: $\frac{\partial p_m(x_i)}{\partial \eta_{ik}} = \frac{0 - \exp(\eta_{im}) \exp(\eta_{ik})}{\left(\sum_\ell \exp(\eta_{i\ell})\right)^2} = -p_m(x_i)p_k(x_i)$.

Combining via the Kronecker delta $\delta_{km}$:

$$\frac{\partial p_m(x_i)}{\partial \eta_{ik}} = p_m(x_i)\left(\delta_{km} - p_k(x_i)\right).$$

Now differentiate the complete log-likelihood $\ell(\theta)$ with respect to $\beta_k$:

$$\begin{aligned}
\frac{\partial \ell(\theta)}{\partial \beta_k} &= \sum_{i=1}^{N}\sum_{m=1}^{K} y_{im}\frac{1}{p_m(x_i)}\frac{\partial p_m(x_i)}{\partial \beta_k} \\
&= \sum_{i=1}^{N}\sum_{m=1}^{K} y_{im}\frac{1}{p_m(x_i)}\left[p_m(x_i)(\delta_{km} - p_k(x_i))x_i\right] \\
&= \sum_{i=1}^{N}\left(\sum_{m=1}^{K} y_{im}\delta_{km} - p_k(x_i)\sum_{m=1}^{K} y_{im}\right)x_i.
\end{aligned}$$

Because $\sum_{m=1}^{K} y_{im}\delta_{km} = y_{ik}$ and $\sum_{m=1}^{K} y_{im} = 1$ (mutually exclusive encoding):

$$\nabla_{\beta_k}\ell(\theta) = \sum_{i=1}^{N}\left(y_{ik} - p_k(x_i;\theta)\right)x_i. \quad \blacksquare$$

**Interpretation:** The gradient vector for class $k$ is an error-weighted superposition of feature inputs:

- The term $(y_{ik} - p_k(x_i))$ represents the empirical prediction residual.
- If sample $i$ truly belongs to class $k$ ($y_{ik} = 1$), the gradient pushes $\beta_k$ along $+x_i$ with magnitude proportional to the underconfidence $(1 - p_k(x_i))$.
- If sample $i$ belongs to a different class ($y_{ik} = 0$), the residual is $-p_k(x_i)$, which pushes $\beta_k$ in the direction $-x_i$ to penalize unwarranted overconfidence.

**Part (c): Hessian Matrix Derivation [6 points]**
To compute the second-order partial derivative block $\mathcal{H}_{k\ell} \equiv \frac{\partial^2 \ell(\theta)}{\partial \beta_k \partial \beta_\ell^T} \in \mathbb{R}^{(p+1)\times(p+1)}$:

$$\frac{\partial^2 \ell(\theta)}{\partial \beta_k \partial \beta_\ell^T} = \frac{\partial}{\partial \beta_\ell^T}\left[\sum_{i=1}^{N}(y_{ik} - p_k(x_i))x_i\right] = -\sum_{i=1}^{N} x_i\left(\frac{\partial p_k(x_i)}{\partial \beta_\ell^T}\right).$$

Applying the chain rule $\frac{\partial p_k(x_i)}{\partial \beta_\ell^T} = \frac{\partial p_k(x_i)}{\partial \eta_{i\ell}}x_i^T = p_k(x_i)(\delta_{k\ell} - p_\ell(x_i))x_i^T$:

$$\mathcal{H}_{k\ell} = -\sum_{i=1}^{N} p_k(x_i;\theta)\left(\delta_{k\ell} - p_\ell(x_i;\theta)\right)x_i x_i^T. \quad \blacksquare$$

**Part (d): Positive Semi-Definiteness and Global Concavity [7 points]**
Let $u = [u_1^T, \ldots, u_K^T]^T \in \mathbb{R}^{K(p+1)}$ be an arbitrary test vector, where each $u_k \in \mathbb{R}^{p+1}$. Examine the quadratic form of the negative Hessian matrix $-\mathcal{H} = -\nabla^2\ell(\theta)$:

$$\begin{aligned}
u^T(-\mathcal{H})u &= \sum_{k=1}^{K}\sum_{\ell=1}^{K} u_k^T\left[\sum_{i=1}^{N} p_k(x_i)(\delta_{k\ell} - p_\ell(x_i))x_i x_i^T\right]u_\ell \\
&= \sum_{i=1}^{N}\sum_{k=1}^{K}\sum_{\ell=1}^{K} p_k(x_i)(\delta_{k\ell} - p_\ell(x_i))(u_k^T x_i)(u_\ell^T x_i).
\end{aligned}$$

Define the real scalar $a_{ik} \equiv u_k^T x_i \in \mathbb{R}$. Expanding the double sum over classes for sample $i$:

$$\sum_{k=1}^{K}\sum_{\ell=1}^{K} p_k(x_i)(\delta_{k\ell} - p_\ell(x_i))a_{ik}a_{i\ell} = \sum_{k=1}^{K} p_k(x_i)a_{ik}^2 - \sum_{k=1}^{K} p_k(x_i)a_{ik}\sum_{\ell=1}^{K} p_\ell(x_i)a_{i\ell}$$
$$= \sum_{k=1}^{K} p_k(x_i)a_{ik}^2 - \left(\sum_{k=1}^{K} p_k(x_i)a_{ik}\right)^2.$$

Notice that since $p_k(x_i) \geq 0$ and $\sum_{k=1}^{K} p_k(x_i) = 1$, the vector $p(x_i)$ defines a valid discrete probability distribution. Therefore, this expression is identically the variance of a discrete random variable $A_i$ taking value $a_{ik}$ with probability $p_k(x_i)$:

$$\sum_{k=1}^{K} p_k(x_i)a_{ik}^2 - \left(\sum_{k=1}^{K} p_k(x_i)a_{ik}\right)^2 = \mathrm{Var}_{Z\sim p(x_i)}(A_i) \geq 0.$$

Because the variance of any real-valued variable is non-negative:

$$u^T(-\mathcal{H})u = \sum_{i=1}^{N} \mathrm{Var}_{Z\sim p(x_i)}(A_i) \geq 0 \quad \forall u \in \mathbb{R}^{K(p+1)}.$$

Thus, $-\mathcal{H} \succeq 0$, which means that $\mathcal{H} = \nabla^2\ell(\theta) \preceq 0$ everywhere in parameter space. **Conclusion:** The log-likelihood function $\ell(\theta)$ is **globally concave**. This guarantees that the optimization landscape is free of sub-optimal local maxima, ensuring that standard first-order (gradient ascent) and second-order (Newton-Raphson / IRLS) algorithms are guaranteed to converge to the unique global maximum likelihood estimate. ■

## Problem 4: Feature Selection & Data Leakage in Cross-Validation [25 points]

**Problem Statement:** Consider a synthetic dataset of $N = 50$ samples and $p = 5000$ independent standard normal predictors $X_{ij} \sim \mathcal{N}(0, 1)$, and binary targets $Y_i \sim$ Bernoulli(0.5) sampled independently. The true generalization Bayes error rate is $\mathrm{Err}_{\mathrm{true}} = 50\%$.

(a) Explain why selecting the top 100 features on the entire $N = 50$ dataset prior to 10-fold cross-validation reports an error rate near 0% (data leakage).

(b) Derive the expected maximum correlation $\mathbb{E}[\max_j |r_j|]$ across 5000 noise features using extreme value theory.

(c) State the correct protocol for cross-validation with supervised feature selection to guarantee zero leakage.

### Solution:

**Part (a): Mechanism of the Flawed Protocol (Data Leakage) [9 points]**
In the flawed validation protocol, the practitioner computes the sample correlation $r_j = \widehat{\mathrm{Corr}}(X_j, Y)$ across all $N = 50$ samples of the dataset **prior** to performing cross-validation splits.

1. **Selection Bias on the Entire Sample:** The $m = 100$ features in $\mathcal{S}$ are selected specifically because their sample correlation with the complete target vector $Y = [Y_1, \ldots, Y_{50}]^T$ is abnormally high.

2. **Leakage into the Validation Holdout:** When 10-fold cross-validation is performed, each fold holds out 5 test samples. However, those 5 test samples have **already contributed** to selecting $\mathcal{S}$!

3. **Artificial Correlation in Test Folds:** Because the features in $\mathcal{S}$ were chosen precisely because their values happened to correlate with the observed $Y$ in those specific 5 holdout observations, the test samples are no longer independent out-of-sample evaluations.

4. **Overoptimistic Reported Performance:** A classifier (such as 1-NN or LDA) trained on the 45 remaining samples will find that these 100 features generalize extraordinarily well to the 5 holdout samples purely because the noise was co-selected. The cross-validation error rate drops to near 0%, fatally misleading the researcher into believing they have found a highly predictive signature when the true generalization rate on genuine unseen data is exactly 50% (random guessing). ■

**Part (b): Extreme Value Statistics of Spurious Correlation [7 points]**
Under the null hypothesis of independence ($X_{ij} \perp Y_i$), the empirical sample correlation coefficient $r_j$ computed on $N = 50$ observations has mean zero and standard error:

$$\mathrm{SE}(r_j) \approx \frac{1}{\sqrt{N}} = \frac{1}{\sqrt{50}} \approx 0.1414.$$

The normalized statistic $Z_j = \frac{r_j}{\mathrm{SE}(r_j)} \approx \sqrt{50}\, r_j$ is asymptotically distributed as standard normal $\mathcal{N}(0,1)$. Because all $p = 5000$ predictors are generated independently, we are observing 5000 independent realizations of $Z_j$.

By the classical Fisher-Tippett-Gnedenko Extreme Value Theorem for Gaussian maxima, the expectation of the maximum order statistic across $p$ independent standard normals is:

$$\mathbb{E}\left[\max_{1\le j\le p} Z_j\right] \approx \sqrt{2\ln p} - \frac{\ln(\ln p) + \ln(4\pi)}{2\sqrt{2\ln p}} \approx \sqrt{2\ln 5000} \approx \sqrt{2\times 8.517} = \sqrt{17.034} \approx 4.127.$$

Translating back to the scale of sample correlations:

$$\mathbb{E}\left[\max_{1\le j\le 5000} |r_j|\right] \approx 4.127 \times \frac{1}{\sqrt{50}} \approx 4.127 \times 0.1414 \approx 0.584.$$

Furthermore, consider the probability that an individual noise feature achieves an absolute sample correlation exceeding $|r_j| \ge 0.35$ (corresponding to $|Z_j| \ge 0.35 \times \sqrt{50} \approx 2.475$):

$$P(|r_j| \ge 0.35) = 2\left(1 - \Phi(2.475)\right) \approx 2 \times 0.00666 = 0.0133.$$

Across $p = 5000$ independent noise features, the expected number of spurious features with $|r_j| \ge 0.35$ is:

$$\mathbb{E}\left[\#\{j : |r_j| \ge 0.35\}\right] = 5000 \times 0.0133 \approx 66.6 \text{ features.}$$

Thus, finding 100 features with sample correlations between 0.35 and 0.60 is virtually guaranteed purely by chance. When $p \gg N$, extreme value statistics guarantees that spurious correlations will mimic real biological or physical signals if feature selection precedes cross-validation splitting. ■

**Part (c): The Correct Cross-Validation Protocol [9 points]**
To ensure completely unbiased model validation, **all supervised preprocessing and feature screening must be embedded strictly inside each cross-validation fold**:

1. **Partition the Data First:** Partition the complete dataset of $N = 50$ observations into $K = 10$ disjoint folds $F_1, \ldots, F_{10}$ of 5 samples each.

2. **Iterate Across Folds ($k = 1, \ldots, 10$):**

   - **Split:** Set aside fold $F_k$ as the holdout validation set $\mathcal{D}_{\text{val}}^{(k)}$ (5 samples). The remaining 9 folds constitute the training set $\mathcal{D}_{\text{train}}^{(-k)} = \mathcal{D} \setminus F_k$ (45 samples).

- **Screen Features on Training Only:** Compute sample correlations $r_j^{(-k)}$ between features and labels using **only** the 45 observations in $\mathcal{D}_{\text{train}}^{(-k)}$.
- **Select Features:** Determine the set $\mathcal{S}_k \subset \{1, \ldots, 5000\}$ of 100 features having the highest $|r_j^{(-k)}|$.
- **Train Classifier:** Fit the classifier (e.g., 1-NN or LDA) using only the subset of features $\mathcal{S}_k$ on $\mathcal{D}_{\text{train}}^{(-k)}$.
- **Validate on Unseen Holdout:** Predict the class labels for the 5 holdout observations in $\mathcal{D}_{\text{val}}^{(k)}$ using the model trained on $\mathcal{S}_k$. Record the validation error rate $\text{Err}_k$.

3. **Aggregate Performance:** Compute the overall cross-validation estimate:

$$\text{CV}_{(10)} = \frac{1}{10} \sum_{k=1}^{10} \text{Err}_k \,.$$

**Theoretical Guarantee:** Because the holdout fold $F_k$ was completely quarantined while selecting $\mathcal{S}_k$, the features in $\mathcal{S}_k$ have zero expected correlation with $Y$ on $F_k$. When this proper protocol is executed on pure noise, $\text{CV}_{(10)}$ correctly yields an error rate of approximately 50%, accurately exposing the lack of true predictive signal. ■